We need to assess both the quantity and the quality of knowledge a person has in those domains of knowledge that are pertinent to the field of work. Of course, it is not easy to define what is pertinent. Many creative achievements have come about precisely because the creator brought to bear some knowledge outside of the usual domain and saw new ways to connect it with domain knowledge well known by most persons competent in the field. These are the kind of connections I refer to as major *integrative reconciliations,* in that the creative person sees how to make the right connections between concepts in two domains of knowledge that were previously regarded as unrelated, or in some cases even contradictory. There may still remain the task of demonstrating to skeptics the power and validity of the new connection(s), but in due course, the dogged persistence of the creative person and his or her adherents is recognized and sometimes rewarded. This kind of intelligence is obviously precious to society at large and to corporations in their quest for competitive advantage. Attention to criteria that indicate emotional intelligence also need to be given more time and effort. Too few of the recruitment criteria used by corporations at the present time recognize that they may be doing a great job of selecting for inert intelligence, but a poor or counterproductive job of selecting for successful intelligence. The latter, from my perspective, is the person who has a history of seeking high levels of meaningful learning and a tenacity to persevere until his/her hunches begin to gain acceptance. Measurement and evaluation issues are discussed further in chapter 9.

It should be evident from the text and figures I presented that I seek to facilitate your subsumption, progressive differentiation, and integrative reconciliation of concepts regarding human learning. This process will be aided if you construct your own concept maps, both of subsets of concepts, and ones for the large groups of concepts. Try to build your own composite maps—and also add illustrative examples from your own experiences. Then refer for comparisons to Fig. 5.1 that shows a composite map for the key ideas of Ausubel's assimilation theory that I have constructed. As you gain in your understanding of assimilation theory, it should be possible for you to gain in your capacity to engage in high levels of meaningful learning. This, I predict, will permit you to be more creative in those domains you choose to pursue.

6

The Nature of Knowledge and How Humans Create Knowledge

THE NATURE AND SOURCES OF KNOWLEDGE

That humans learn is self-evident. It is also self-evident that humans organize and communicate knowledge to one another. What is not obvious is the origin of knowledge. Where does knowledge come from? This has been a question pondered by some of the best minds for centuries. Most of the great philosophers throughout history have spoken and written on this question.

It is not my purpose to review the long history of philosophical ideas about the nature of knowledge and knowing, but it is necessary to deal with some of the ideas that have been dominant in the past 300 years because they continue to influence teaching, learning, schools, businesses, and society today. First, however, I try to make clear the philosophical ideas that now guide our work and my answer to the question: Where does knowledge come from? The branch of philosophy that deals with the structure and origins of knowledge is called *epistemology,* and I shall try to make clear my epistemological ideas.

All knowledge is comprised of concepts and propositions, including concepts and propositions that deal with learning strategies and methods of conducting inquiries and also including the affective dimension of experience associated with those concepts and propositions. Meaningful learning underlies the constructive integration of thinking, feeling, and acting that occurs in human learning and in new knowledge construction. This interplay is unique to human beings and hence I choose to label it *human constructivism* (Novak, 1993). Human constructivism is a label I see as appropriate both for the way in which humans learn their usable

knowledge and also for the way in which they construct new knowledge. The nature and process of meaningful learning underlies both human learning and human knowledge creation.

I next discuss some older ideas about the nature of knowledge and origins of knowledge and reasons why I believe these ideas are in error, less powerful, or less relevant to education and knowledge creation. In this chapter I seek to present a view of epistemology that I believe has power for improving both. Whether it has power for the advance of the field of philosophy dealing with epistemology is for others to decide. It will be seen that ideas of human constructivism are complementary to or congruent with emerging ideas on epistemology being advanced by contemporary philosophers and historians interested in the process of knowledge production.

Gowin's Epistemological Vee

My colleague, D. Bob Gowin (now retired), has been interested in the study of philosophy and epistemology as it applies to education, beginning with his Master's degree work 4 decades ago. His interests and expertise in philosophy have been valuable to our research program in science education, especially in research studies that have focused on science laboratory learning. In general, most students experience considerable anxiety and confusion in science laboratory work, especially when this work involves experimental or quasi-experimental studies. College students also find the reading of research reports frustrating or confusing, and sometimes completely opaque.

To aid students in understanding research reports, Gowin (1970) devised five questions, the answers to which could provide the student with a better understanding of the research. These five questions were:

1. What are the telling questions? These are questions that tell what the inquiry seeks to find out.
2. What are the key concepts? These are the dozen or so disciplinary concepts that are needed to understand the inquiry.
3. What methods of inquiry (procedural commitments) are used? These are the data gathering or data interpreting methods used.
4. What are the major knowledge claims? These are the answers claimed by the researcher as valid answers to the telling questions.
5. What are the value claims? These are claims, explicit or implied, about the worth or value of the inquiry and the answers found in the inquiry.

Gowin and I found the use of these questions to be helpful to students not only in the analysis of research reports but also in laboratory work, for the design of

research, and as a tool for discussions on the meaning and value of research studies. However, many students found it difficult to relate key concepts to the telling questions or the events or objects being investigated. In pondering the problems experienced, Gowin came up with the idea of the Knowledge Vee heuristic in early 1977. Figure 6.1 shows the general form of the Vee.

We have found it is relatively easy to teach persons to construct Vee diagrams once they are familiar with concept mapping and the ideas that underlie concept maps. Appendix 2 shows a set of procedures we have found to be effective. The ideas presented in this chapter will aid in understanding both the concept map

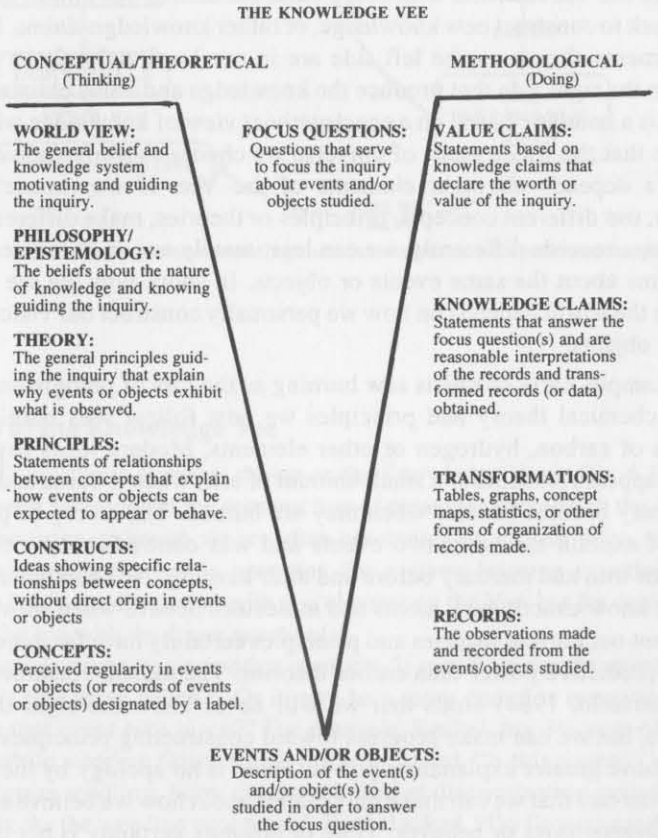

FIG. 6.1. The Knowledge Vee, showing the relationships between epistemological elements involved. All elements interact with each other in knowledge creation. The conceptual/theoretical elements play a central role in selecting events and/or objects to observe, developing focus questions and guiding the methodological activities to construct knowledge and value claims.

tool and the Knowledge Vee tool.

The beauty of the Vee heuristic is its comprehensiveness and also simplicity. It serves to illustrate that there are a dozen or so epistemic elements that are involved in constructing or examining a piece of knowledge, and yet it places these elements into a simple structure that helps to illustrate how each of these elements function. Although a variety of shapes for an epistemological heuristic could have been used, Gowin chose the Vee shape because it points to events/objects, that segment of the universe we are trying to understand. It also serves to distinguish the fact that both thinking (conceptual/theoretical) elements and doing (methodological) elements are involved in knowledge construction. Each element on the Vee interacts with every other element as our minds, hearts, and bodies work to construct new knowledge, or rather knowledge claims. However, those elements shown on the left side are in our head and help to guide the actions on the right side that produce the knowledge and value claims. The Vee heuristic is a heuristic based on a constructivist view of knowledge wherein we recognize that the way a piece of universe we choose to study appears to look or behave depends on other elements of the Vee. If we choose different questions, use different concepts, principles or theories, make different records or transform records differently, we can legitimately arrive at different knowledge claims about the same events or objects. In short, how we see events or objects in the world depends on how we personally construct our vision of these events or objects.

For example, early chemists saw burning as the loss of phlogiston, whereas the new chemical theory and principles we now follow sees burning as the oxidation of carbon, hydrogen or other elements. Modern ideas explain why wood disappears (except for a small amount of ash) when it burns and why iron and mercury become heavier when they are burned. The theory of phlogiston could not explain the latter two events and was contradicted by the weight records for iron and mercury before and after burning. No chemist or physicist claims to know exactly how atoms and molecules behave when substances are burned, but our current theories and principles certainly have far more explanatory and predictive power than earlier theories. The radical constructivist (see von Glasersfeld, 1984) holds that we will never know the right theories or principles, but we can make progress toward constructing principles and theories that have greater explanatory power. There is no apology by the constructivist for the fact that we can make only claims about how we believe some piece of the universe looks or behaves. Truth or absolute certainty is not the goal of the constructivist thinker. It has been the goal of positivist/empiricist philosophy for hundreds of years, and positivistic views abound in textbooks, lectures, and schooling in general. I return to this issue later in this chapter. Figure 6.2 shows some of the distinctions between constructivist and positivist or empirical epistemologies.

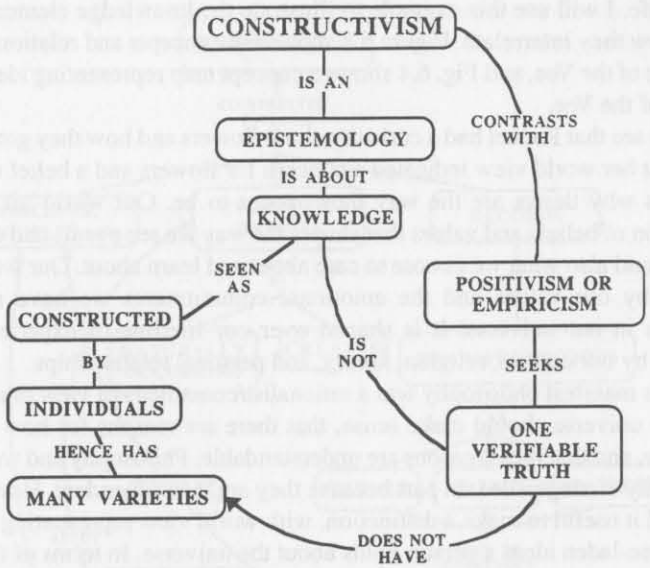

FIG. 6.2. Some of the key concepts of constructivism contrasted with positivism/empiricism.

Elements of the Knowledge Vee

There are 12 epistemic elements shown on the Knowledge Vee in Fig. 6.1. Each of these elements functions in the construction of meanings and also in the construction of knowledge, although we are often not conscious of their roles, nor do we consider how each element is operating for a given learning experience. It is possible to begin the discussion with any element on the Vee, but for convenience, I begin at the top with the focus question(s).

Often new learning begins with a question. It may be a simple question such as, "what is that thing called?" Or it may be a more complex question such as, "how does that event take place?" For example, Rachel, my 3¾-year-old granddaughter, while weeding flowers with my wife, asked, "Is this a weed?" pointing to a small grass seedling. Soon she was skillful at discriminating young flowers from weeds. As the weeding progressed, Rachel asked, "Do flowers eat dirt?" My wife explained that flowers do use some of the dirt to grow, but mostly they used water, air, and energy from the sun. Rachel then asked, "When the flowers get big, will there be less dirt?" to which my wife responded, "Yes, but only a very tiny bit less." What is evident in Rachel's questions was that she was not operating in a conceptual/theoretical vacuum; she had some concepts, principles, and theories about what plants are and how they grow, acquired over the last 2 or 3 years of

her short life. I will use this example to illustrate the knowledge elements of the Vee and how they interrelate. Figure 6.3 shows the concepts and relationships for the left side of the Vee, and Fig. 6.4 shows a concept map representing ideas on the right side of the Vee.

First we see that Rachel had a curiosity about flowers and how they grow. It was evident that her world view indicated a concern for flowers and a belief that there are reasons why things are the way they appear to be. Our world view is that constellation of beliefs and values that shapes the way we see events and objects in the world, and also what we choose to care about and learn about. Our world view is shaped by our values and the emotional commitments we have regarding happenings in our universe. It is shaped over our lifetime of experiences and influenced by our culture, religion, family, and personal relationships.

Rachel's manifest philosophy was a rationalist/constructivist view that happenings in the universe should make sense, that there are reasons for how and why plants grow, and that these reasons are understandable. Philosophy and world view are not easily distinguished, in part because they are interdependent. However, we have found it useful to make a distinction, with world view representing the more global, value-laden ideas a person holds about the universe. In terms of the Vee, it

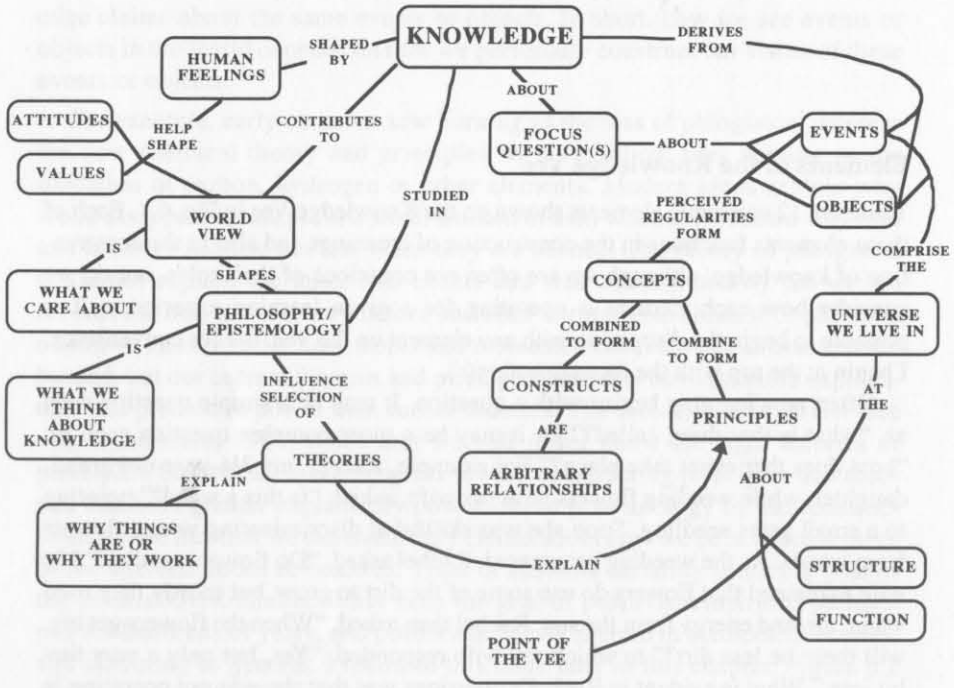

FIG. 6.3. A concept map representing the ideas involved in the elements on the "left side" of the Knowledge Vee.

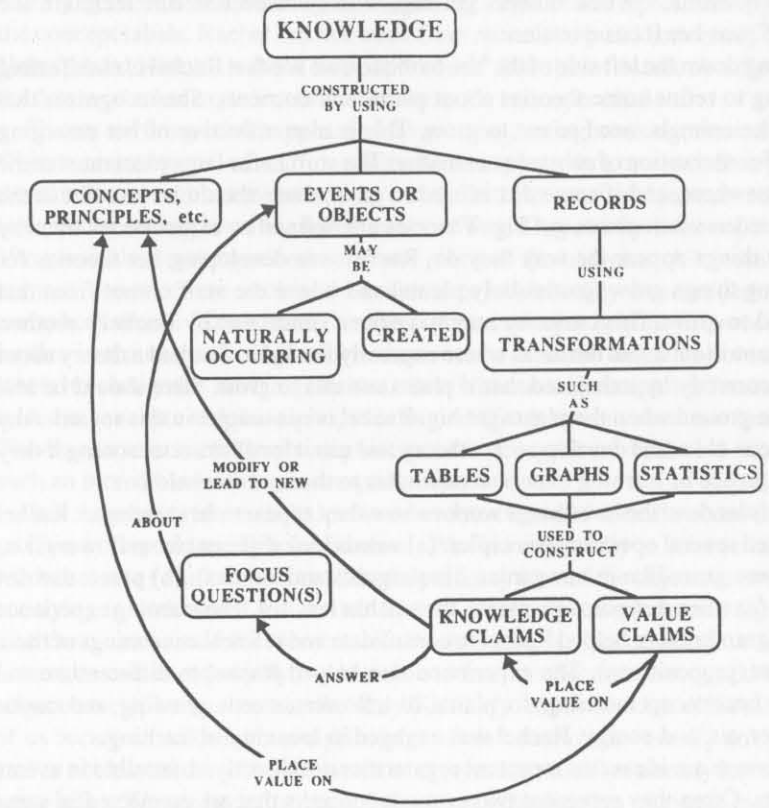

FIG. 6.4. A concept map representing the ideas involved on the "right side" of the Knowledge Vee.

is best to think of philosophy in terms of our epistemological beliefs, that is, where do we believe knowledge comes from?

Matthews (1980, 1984) did an analysis of interviews where young children (6- to 8-years old) raised some of the same questions and issues and presented similar arguments as those recorded in the writings of some of the greatest Western philosophers. Even very young children have already established philosophical ideas about how people behave and how the world works. In this respect, Matthews (1980) rejected outright Piaget's claim that philosophical reasoning does not develop until the teenage years.

Our world view motivates us to act, to construct questions, and to find answers. We have chosen to place focus question(s) at the top center of the Vee because these, in many ways, are what drive the inquiry that leads eventually to new knowledge.

Rachel's question, "When flowers get big, will there be less dirt there (in the garden)?" was her focus question.

Moving down the left side of the Vee to theory, we see that Rachel is manifesting and trying to refine some theories about plants and nutrients. She recognizes that plants, like animals, need to eat to grow. This is also reflective of her emerging theory of conservation of substance or matter: The stuff in the large plant must come from somewhere, and if some dirt is used to grow, there should be less dirt in the flower garden when plants get big. Theories are defined as explanations for why and how things appear the way they do. Rachel was developing her theories for how living things grow (particularly plants) and where the stuff comes from that they need to grow. This curiosity may have been stimulated by Rachel's mother, who is a nutritionist, but the ideas where expressly Rachel's. She had a theory about dirt and correctly hypothesized that if plants use dirt to grow, there should be less dirt in the ground when the plants get big. Rachel is not unique in this regard. Any 3- or 4-year-old could develop such a theory and use it for abstract reasoning if they had a sequence of learning experiences similar to those of Rachel.

Principles describe how things work or how they appear to be structured. Rachel manifested several operating principles: (a) weeds look different from flowers (i.e., weeds were grass-like in our garden of marigolds and zinnias); (b) plants use dirt to grow; (c) when dirt goes into plants, the soil has less dirt. The learning experience with her grandmother helped Rachel to consolidate and refine the meanings of these principles (propositions). The experience also helped Rachel to differentiate and integrate her concept meanings for plants, dirt, flowers, weeds, growing, and maybe also water, air, and energy. Rachel was engaged in meaningful learning.

Constructs are ideas that represent regularities nor directly observable in events or objects. Often they represent two or more concepts that are connected in some arbitrary way. For example, if Rachel were to suggest that her grandmother's flowers were growing in healthy air or healthy dirt, she would be using constructs. If Rachel had concepts of healthy and some concept of air and dirt, she could combine these to suggest the conditions under which the flowers were growing. Halpern (1989) defined *hypothetical constructs* as concepts having no external reality, giving learning and memory as examples. This could also be said of atoms, love, and photosynthesis, but it seems much more coherent to label these as concepts, representing specific regularities in events or objects, albeit they may be observed indirectly. Constructs such as IQ are arbitrary relationships between concepts—there is no necessary reason to divide mental age by chronological age. Arbitrary constructs are more common in the social sciences. Constructs differ from principles because they do not explain how some aspects of the universe functions or is structured.

I have defined *concept* as a perceived regularity in events or objects, or records of events or objects, designated by a label. Clearly Rachel had established meanings for the concept labels just noted, and was refining these meanings; that is, she was

becoming more discriminating as regards the regularities represented by each of the concept labels. Rachel did not work from records to build her concepts, as Van Helmont did with his willow tree experiment,[1] but she will undoubtedly do so in the future in other experiences that relate to the concepts cited.

Events are things that happen such as growing, eating, running, wars, and so on. Objects are some unit of stuff or matter, such as plant, dirt, or weed. Everything in the universe is either an event or object, and all events involve objects; even changes in forms of energy involve objects in some way. Thus, at the bottom point of the Vee we anchor our experience to some piece of the universe. All meanings that humans construct are anchored in events or objects they have experienced, or metaphors drawn from events or objects. Our concepts help us to perceive regularities in the events or objects we observe, and in some cases we construct new labels to designate new regularities. But without a functioning framework of relevant concepts and principles, it is more difficult to construct new knowledge. This is one reason children's acquisition of language from birth to 3 years old is such an incredible learning accomplishment!

It is very important to help learners become clear and specific about the events or objects they are trying to understand. We find repeatedly in science laboratory work that many students have at best only a fuzzy idea of the objects or events they are trying to understand and for which they are trying to seek out the regularities. The same can be said for elements of events in sports, dance, music, or literature. Literature can be especially troublesome to learners because most good literature relies on metaphor to structure the story. Similarly, mathematics is difficult for most of us because the concepts and principles of mathematics are usually not specified and/or not related to real world events or objects for which we already have relevant knowledge. We see later that lack of clear conceptualization of events in organizations is also a problem.

Rachel's questions signaled clearly what events and objects she was trying to understand. Rachel was engaged in a relatively high level of meaningful learning. We could follow through with examples of how elements on the right side of the Vee could function in the construction of knowledge and value claims, but the limited context of Rachel's learning did not include making records or transforming records. Examples of this will come later in her life.

Experiments

Emerging in the 17th century, Galileo and others we now called scientists (they saw themselves as natural philosophers) developed the experiment. Experiments are

[1]In 1650, Van Helmont inquired into the source of materials needed for plant growth. He grew a willow twig in a tub of soil and concluded (erroneously) that all the substance used to grow came from water. He did not know about the composition of air, and carbon dioxide as a gas in the air was not known until much later. His scales could not detect the small amount of nutrient minerals absorbed from the soil.

events created by researchers, and the conventional experiment requires that we observe both experimental events and control events. The requirement includes that all elements we are observing should be identical, except the one experimental variable. Differences in records we obtain from the events permit us to test hypotheses regarding the influence of the experimental variable on the event being studied. How we record and transform the records is determined, in part, by the hypotheses we seek to test. Hypotheses could also be called "anticipated knowledge claims." In the 20th century, new statistical tools were developed that permit experiments to be done with multiple variables or experimental conditions, thus also permitting construction of hypotheses about how two or more variables interact to produce the records we obtain.

The success of experimental procedures in the natural or hard sciences as ways of producing useful knowledge has led to widespread efforts to apply this methodology to the social or soft sciences, including education. The two major difficulties in the social sciences are that we can seldom truly control events that involve people, and secondly, our methods of making records have serious problems of validity and reliability (see later this chapter and chapter 9). However, used with sufficient caution and guided by theory and principles that have reasonable validity, experiments can prove useful for knowledge creation in the social sciences as well.

Records

These are literally the records we gather about events or objects (see also chapter 9). They can be simple descriptions of observations, such as the number of each kind of object observed, or they may be meter readings, computer print-outs, or other records made by complex instruments. In the latter cases, there is always the issue of reliability and validity of the records. Faulty equipment or the improper selection of equipment can produce faulty records. *Fact* is a label I use for accurate, valid records, and many records gathered in research are not facts, especially in educational research where the data gathering instruments (e.g., tests) are never highly reliable and highly valid record-making tools. Unfortunately, it is not always easy to know when our records are truly facts, acceptable but containing some error, or biased and distorted. Learners need to be helped in ways to make validity checks and reliability checks on their data. In work settings, workers often know better than their managers the limitations on records gathered. Corporations that respect and use this knowledge enhance their productivity.

Many advances in the sciences have come from the invention of new ways to make records of events. The telescope and the oscilloscope are two examples. Telescopes have led to enormous advances in astronomy, and oscilloscopes have advanced our understanding of electricity and electromagnetic waves. Much of what we call technological advances results from modifying instruments developed for basic research to practical uses. Oscilloscopes laid the groundwork for television, and concept maps are central to our programs to help learners to learn and to

create knowledge. Some research groups become so successful creating events or in refining record-making tools that they lead the world in new knowledge creation in their specialty. For example, this is so true for the research group Darmstaat, who have become so skillful in creating new elements that they are credited with creating elements, 107, 108, 109, and most recently, element 110 (Clery, 1994).

The records we choose to make depend on the question(s) we hope to answer, and all of the elements on the left side of the Vee. Usually we focus our attention on the principles guiding our inquiry because these describe regularities or relationships we may find in the objects or events we are studying. We need to be sure that we are gathering records that are consistent with our principles and may allow us to confirm or deny the validity of the principles guiding our inquiry.

Artifacts

Artifacts are the records of human activity. The implements, pottery, and jewelry studied by archaeologists are products that are used to reconstruct the life of prehistoric people. Artifacts do not occur naturally in the universe but depend on human thinking and activity. Because humans have an infinite capacity to change their minds, artifacts as records can give different messages from different people or from different times. There are regularities in artifacts, and reliable predictions can be made using this kind of record. Much that we deal with in education and all of the social sciences is basically artifactual records such as test scores, interview data, and opinions or feelings about things. Although it is always a slippery business to interpret records of events or objects, it is especially difficult to interpret artifactual records. This is one of the reasons the social sciences are less advanced than the natural sciences. It is also a reason why viable theory may be even more powerful in the social sciences as a tool for guiding action. Theory helps us make judgments as to whether or not the artifactual records we used and the claims we constructed have a reasonable chance of being valid.

Record Transformations

Usually we do not try to construct knowledge claims from raw data or records as they are gathered from our observations. It is common to do some kind of transformation of the new data. Simple groupings, tables, charts, and graphs are some of the common record transformations we use. *Data* is a term usually used for any records or transformed records. Although Rachel did not record in writing her observations of flowers, she was making mental records of her observations and using previous records she had observed.

Record transformations we make should be guided by our concepts, principles, and theories. The transformations are also determined by the focus question(s) we hope to answer. Principles help us to organize our data to show patterns or relationships anticipated by applying our principles. For example, there is a

principle of economics that holds that interest rates are dependent on the money supply in an economy. Thus, applying this principle, we would have gathered records on money supply (as reported from federal sources) each week or month for some period of time, and also records on the interest rates for each week or month. We could place these records into a table showing dates in one column, money supply in another column, and interest rates in a third. However, a better view of the relationship would be a graph showing money supply on one axis and interest rates on the other. A nice line graph showing steady increase in interest rates with a regular decline in money supply would confirm or support our economic principle. Alas, if this did not occur in our record transformation, we might question the validity of the principle. More commonly, we begin to recognize that something else must be happening, and we may reach into our store of economic principles to see if any other principles can explain our graph. We may find the principle that consumer buying is related to interest rates, and now we must go back and gather more records, records on consumer spending over the time interval we are studying. Then we construct new transformations of our records. Of course, it is always possible that our principles may be faulty, and economics is a good example of a field where there is considerable dispute among experts as to which principles are valid or which are most important. For one thing, we are dealing with events (e.g., consumer willingness to buy or to borrow money) that are choices people make. The records of interest rates are therefore artifacts, not facts.

Statistical record transformations are very common in many fields, especially in educational research. Unfortunately, statistical transformations will not improve biased or invalid records, and these are all too common in the social sciences. Statistics will not make facts out of records that are artifacts. Statistical tests and their interpretation are also subject to sets of concepts and principles, and too often these are not known or ignored by the researcher. For a good critique of statistics in education, especially the common use of factor analysis and similar tools, see Gould (1981).

Knowledge Claims

These are the answers to the focus questions that we claim our records and record transformations lead us to. Or we may simply make a knowledge claim from our observations of events without the precision that may come from good records and record transformations. Rachel's conjecture that there would be less dirt in the garden when the plants got large would be an example of this. The commonly used term, *hypothesis,* is an anticipated knowledge claim. However, hypotheses are statements we are trying to prove or falsify. From a constructivist perspective, hypotheses have only limited value. Gathering different records or using different record transformations could lead to very different answers to the same questions about the same events. Moreover, we can never be certain about the validity or reliability of our records, and at least to some degree, deficiencies in our records

could lead to faulty knowledge claims. As we noted earlier, applying different concepts, principles, or theories could change entirely the knowledge claims that emerge in any inquiry. Constructing knowledge claims is indeed a slippery business. There are so many ways to go wrong, so many ways to be in error. This is especially true in any field where principles and theories are either lacking or of dubious validity. This has been the case in much educational research (as well as research in other social sciences), so there is good reason why teachers and the general public are skeptical about so called educational research findings. Moreover, with poor or nonexistent theory, questionable principles, and limited record making tools so common in educational research, it is no wonder that research findings are often contradictory. The most common finding when alternative teaching or learning strategies are compared is: "There were no statistically significant differences between groups (or methods)." Kaestle (1993) commented on the awful reputation of educational research. He noted a lack of influence of research on practitioners, the disarray of the educational community, and the politicization of the field. He failed to note the theoretical poverty of the field.

Value Claims

These are claims about the value or worth of the inquiry for achieving the goal(s) that motivated the study. For example, some of our studies were designed to ascertain whether or not concept maps and Vee diagrams could be helpful to students, both in terms of knowledge achievement and in terms of gains in self confidence or interest in the field of study. Our data generally support the knowledge claim that concepts maps and Vee diagrams help learners to learn and to gain in confidence and interest. (See, e.g., Novak, Gowin, & Johansen, 1983.) Therefore we make the value claim that these are useful tools that should be used by teachers and learners. Value claims are always linked to knowledge claims, but they are not the same thing. We have found it helpful to students and teachers to make deliberate efforts to identify and record value claims for every inquiry. This also helps both students and teachers to recognize that knowledge construction is a very value-laden endeavor! If Rachel had exclaimed that it was wonderful that beautiful flowers could grow from air, water, and dirt, she would have stated a value claim about flowers. I believe most of us would agree with such a value claim.

Our world view and our philosophy will, in many ways, influence the fields we choose to study and the kinds of questions we seek to answer. There is a strong relationship between the value claims we seek to construct and our philosophy and world view. An important component of my world view is that I believe human beings have the power to reduce the degree of human suffering in the world, and that improved education, based on theory, is an important endeavor to reduce human suffering. That is why I have chosen education as a field of work, and why I have been striving to develop a theory and related principles to guide our research and practice. I may not live long enough to see convincing evidence that theory-based

improvement in education can reduce human suffering, but some of our more recent work, and the work of others, makes me optimistic.

There is an important difference between schools or universities and the business world. Today every large business faces fierce competitive pressures, not only from other businesses in this country, but also from businesses located all over the world. The globalization of business, as so many writers are pointing out, means that survival of the business requires rapid, effective, and efficient new learning and new knowledge creation. Nonaka and Takeuchi's (1995) assertion that all businesses must become knowledge creating organizations suggests to me that the ideas and tools I am presenting in this chapter may find more rapid acceptance and application in the business world than in the academic world. They do not have the luxury of continually increasing taxpayers' support to sustain their ineffective approaches.

There are other epistemic elements that could be discussed; however, this is not a book on epistemology. Nevertheless, I regard an understanding of epistemology as essential to understanding the nature of the knowledge we seek to teach or to learn. It is essential for teachers, learners, and managers to acquire *metaknowledge,* that is, knowledge about knowledge. Of course, one could argue that teachers have taught and learners have learned and managers have managed for centuries without understanding the nature of knowledge and the processes involved in knowledge construction. The point is, if we want to improve the educative process by a quantum leap, teachers, learners, and managers need to learn more, not only about how humans learn but also about how they create knowledge. This is the central goal in our book, *Learning How To Learn* (Novak & Gowin, 1984).

We have found the Vee heuristic to be helpful to students in our research group for designing their own research projects, and also to serve as a vehicle for dialogue between individuals. On a single page, it is possible to present the key elements that are guiding the inquiry, the objects or events to be studied, the focus questions to be pursued, and the elements that will be used to construct knowledge and value claims. Figure 6.5 shows an example of a Vee constructed by one of my former PhD students for her thesis work in nutrition education. She has gone on to achieve a worldwide leadership position in the field, partly as a result of the power of her theory-driven research program.

The Interrelationship of Learning and Knowledge Creation

Throughout the last chapter and in this chapter, I have tried to show how I see meaningful learning and knowledge construction as highly related. In fact, in my view all knowledge construction is only an extension of the human capacity to construct new meanings (new concepts and concept relationships in cognitive structure). Thus, the psychology of meaningful learning underlies and gives rise to the epistemological process of knowledge construction. Human Constructivism is

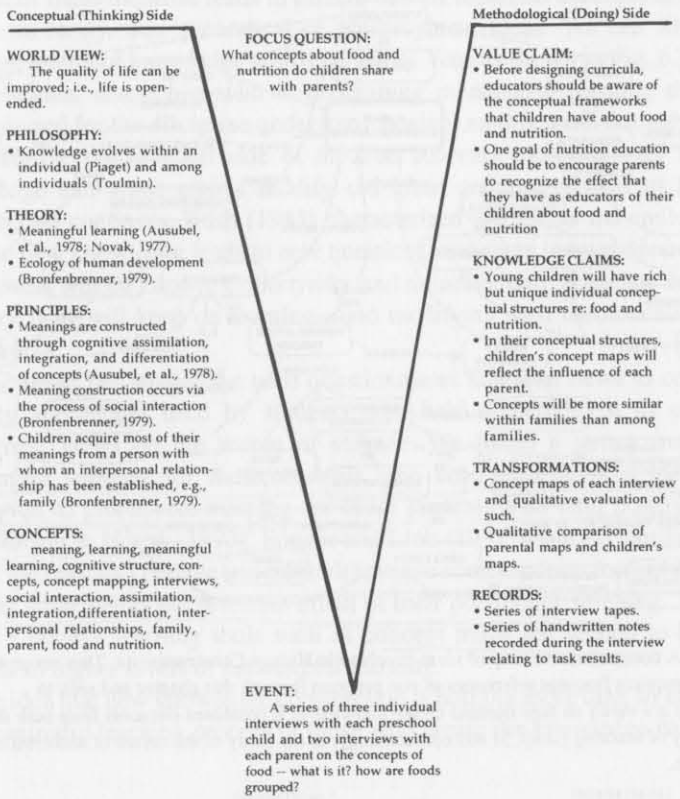

FIG. 6.5. A vee heuristic that (a) defines each element involved in the process of knowledge construction (in italics) and (b) illustrates the plan for a theory-driven inquiry in nutrition education. "Theory Driven Research as a Means to Improve Nutrition Education," by C. L. Achterberg, J. D. Novak, and A. H. Gillespie. Reprinted with permission, *Journal of Nutrition Education, 17*(5), p. 180, 1985, Society for Nutrition Education.

both a psychological and an epistemological phenomenon. This view is illustrated in the concept map in Fig. 6.6.

Figure 6.6 is an exceedingly complex figure. Do not expect to understand all of the concepts and relationships represented in it with a single reading. Instead, use it as a reference map to see how ideas presented in this chapter and chapter 5 interrelate with one another. Most of my graduate students find it takes them 1 to 3 years to feel comfortable with the ideas of human constructivism, including some time to shake off old ideas and attitudes that are positivistic in character. Our society is permeated with positivistic thinking, and it is not easy for one to move away from this. Behaviorist theories of learning may be almost dead in the world of psychology, but they are alive and well in schools and corporations. Neither positivist